

The Knowledge Lifecycle of Large Language Models

A model holds two memories. When they disagree, training usually wins.



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The problem

How a language model handles a fact is studied by **five separate communities** that rarely cite one another.

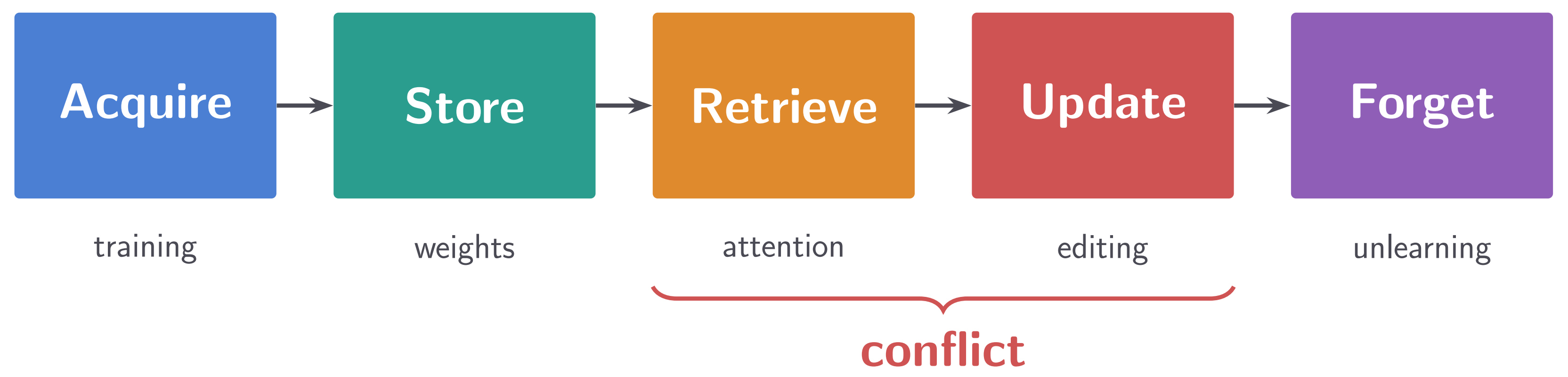
- retrieval-augmented generation
- knowledge editing
- continual learning
- context engineering
- agent memory

Each has its own benchmarks, and each passes them. A system can pass **all five** and still fail, because nothing tests what happens **between** them.

Across **20 representative papers**, the median work covers **two** of the five stages. The boundaries are where deployment breaks.

The framework

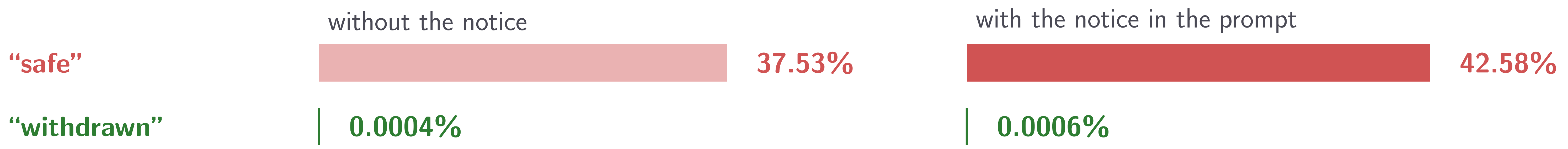
A fact passes through five stages. Naming them as one lifecycle is what makes the boundaries between them visible.



The failure below sits between **Retrieve** and **Update**: the document arrives, and nothing tells the model which copy of the fact to trust. That boundary belongs to neither field beside it, which is why nobody tests it.

The finding

Vioxx was withdrawn worldwide in September 2004 for raising the risk of heart attack and stroke. Put that withdrawal notice **directly into GPT-2's prompt**, then ask whether the drug is safe to prescribe.



Bars are true to scale. The correct answer's bar would be 1/72,582 the width of the bar above it, so a tick marks where it begins.

The correction is sitting in the prompt, and confidence that the drug is safe goes up.
Retrieval worked. What failed is the step after it, where the model has to decide which of its two memories to believe.

A metric

Lifecycle Desynchronization is the surprisal of the **correct** answer while the corrective document is present:

$$\hat{\mathcal{D}}_{sync} = -\ln P(y^* \mid q, \theta_{stale}, \mathcal{M}_{now})$$

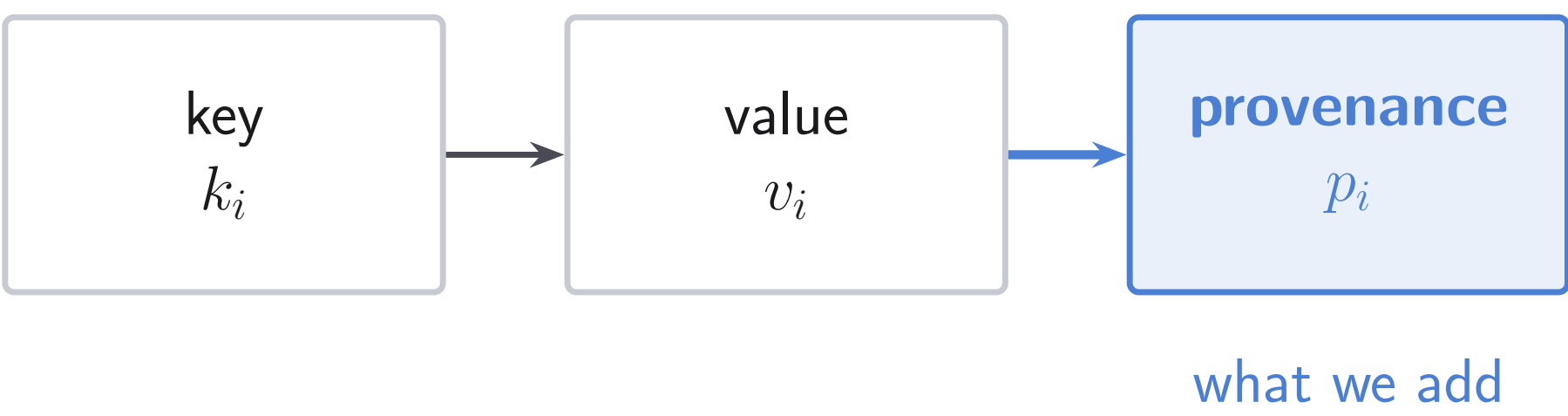
It is read in nats: n nats means probability e^{-n} . Past **9.2** the correct answer is below one chance in ten thousand.



A companion number, **context influence**, asks whether the document moved the model at all. Here it is **0.033** nats: the evidence is present, and **inert**. The notice is placed in the prompt by construction, so nothing was lost in retrieval.

An architecture

Storage records **what** is true but never **when** it was learned, so nothing lets the model prefer fresh evidence over a confident old memory.



The **Provenance Vector** attaches acquisition time to every memory slot. A gate reads it at inference and mutes facts that have gone stale:

$$S(\Delta\tau_i) = \sigma(\gamma - \beta \Delta\tau_i)$$

Weights are never modified, so the specificity bottleneck that limits static editing does not apply. Under **0.02%** extra parameters.

Reproduce it

Every number on this poster comes from **one deterministic script**. Nothing is sampled, so the digits come out identical on any machine.

```
pip install torch transformers
python dsync_experiment.py
```

It runs on **GPT-2 base**, 124M parameters: small, fully open, and with no instruction tuning to confound the result. A laptop is enough.

The metric is **measured**. The architecture is **proposed** and has not been trained.

One model, one fact, one prompt. The size of the effect is what needs replication, and the repository carries the notebook for doing it.

Paper and code

github.com/Amey-Thakur/LLM-KNOWLEDGE-LIFECYCLE

Live demonstration

huggingface.co/spaces/ameythakur/llm-knowledge-lifecycle

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